LabCorp
Laboratory Corporation of America

UNIVERSITY PATHOLOGISTS, P.C.
University of Tennessee Medical Center

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Drawer U-108
1924 Alcoa Highway
Knoxville, TN 37920

Surgical Pathology Report

Patient: [redacted]
Med. Rec. #:
DOB/Gender:
Soc. Sec. #:
Location:
Physician(s):

[redacted]

Specimen(s) Received

A:Right breast
B:Additional anterior margin at right nipple base
C:Left breast
D:Additional anterior margin at left nipple base
E:Sentinel lymph node #1, left axilla
F:Sentinel lymph node #2, left axilla
G:Sentinel lymph node #3, left axilla

Clinical History and Diagnosis

Left IDC, multifocal, three clips

Final Pathologic Diagnosis

A) BREAST, RIGHT, SKIN-SPARING TOTAL MASTECTOMY:
1) SCATTERED AREAS OF FLAT EPITHELIAL ATYPIA
2) PROMINENT FIBROCYSTIC CHANGE WITH A LARGE 15 mm CYST CONTAINING DENSE INSPISSATED SECRETIONS LOCATED WITHIN THE UPPER INNER/CENTRAL QUADRANT (SEE COMMENT)
3) INTRADUCTAL PAPILLOMA
4) AREAS OF PROMINENT ADENOSIS AND SCLEROSING ADENOSIS
5) AREAS OF USUAL DUCTAL HYPERPLASIA
6) STROMAL FIBROSIS
7) SMALL INTRAPARENCHYMAL LYMPH NODE, NEGATIVE FOR MALIGNANCY (0/1)
8) NEGATIVE FOR DUCTAL CARCINOMA IN SITU AND INVASIVE CARCINOMA

B) BREAST, RIGHT, ADDITIONAL ANTERIOR MARGIN AT NIPPLE BASE, EXCISION:
NEGATIVE FOR DUCTAL CARCINOMA IN SITU AND INVASIVE CARCINOMA

C) BREAST, LEFT, SKIN-SPARING TOTAL MASTECTOMY:
INVASIVE MAMMARY CARCINOMA, DUCTAL TYPE, NOT OTHERWISE SPECIFIED
- NOTTINGHAM SCORE: 6 OF 9 (TUBULES = 2; NUCLEI = 2; MITOSES = 2)
- HISTOLOGIC GRADE: MODERATELY DIFFERENTIATED (G2)
- TUMOR SIZE: NUMEROUS VARYING SIZED FOCI OF INVASIVE CARCINOMA ARISING

IN CLOSE PROXIMITY TO ONE ANOTHER THAT COALESCE TOGETHER TO FORM A SINGLE LARGE CONFLUENT TUMOR MASS SPANNING 39 MM IN GREATEST DIMENSION THAT IS LOCATED WITHIN THE LOWER OUTER/CENTRAL BREAST SPECIMEN AT APPROXIMATELY 4-5 O CLOCK
- LYMPH VASCULAR INVASION: NOT IDENTIFIED
- PERINEURAL INVASION: PRESENT
- IN SITU COMPONENT: PRESENT, DUCTAL CARCINOMA IN SITU
  NUCLEAR GRADE: 2 OF 3
  TYPES: MICROPAPILLARY AND PAPILLARY TYPES
  SIZE/EXTENT: ASSOCIATED WITH THE INVASIVE CARCINOMA
- MARGINS: NEGATIVE FOR INVASIVE CARCINOMA AND CARCINOMA IN SITU WITH THE CLOSEST MARGIN BEING POSTERIOR WITH ONLY A COUPLE OF FIBROBLASTS PRESENT BETWEEN THE INKED MARGIN AND INVASIVE CARCINOMA. INVASIVE DUCTAL CARCINOMA AND DUCTAL CARCINOMA IN SITU ALSO COMES EXCEEDINGLY CLOSE TO THE ANTERIOR MARGIN AT LESS THAN 1 mm

ADDITIONAL PATHOLOGIC FINDINGS:
- THREE (3) METAL BIOPSY CLIPS WITH ASSOCIATED PRIOR BIOPSY CHANGES
- MULTIPLE SMALL FIBROADENOMAS
- EXTENSIVE FIBROCYSTIC CHANGE
- ADENOSIS AND SCLEROSING ADENOSIS
- PROMINENT STROMAL FIBROSIS

D) BREAST, LEFT, ADDITIONAL ANTERIOR NIPPLE-BASE MARGIN, EXCISION:
NEGATIVE FOR DUCTAL CARCINOMA IN SITU OR INVASIVE CARCINOMA

E) LYMPH NODE, LEFT AXILLA, SENTINEL #1, EXCISION:
ONE LYMPH NODE POSITIVE FOR METASTATIC CARCINOMA (1/1)
- SIZE OF LARGEST METASTATIC FOCUS: 2.1 mm
- EXTRACAPSULAR EXTENSION: NOT IDENTIFIED

F) LYMPH NODE, LEFT AXILLA, SENTINEL #2, EXCISION:
THREE LYMPH NODES, NEGATIVE FOR MALIGNANCY BY ROUTINE H&E AND PANCYTOKERATIN IMMUNOHISTOCHEMISTRY (0/3)

G) LYMPH NODE, LEFT AXILLA, SENTINEL #3, EXCISION:
ONE LYMPH NODE NEGATIVE FOR MALIGNANCY BY ROUTINE H&E AND PANCYTOKERATIN IMMUNOHISTOCHEMISTRY (0/1)

SYNOPTIC SUMMARY (LEFT BREAST)

PATHOLOGIC STAGING
PRIMARY TUMOR: pT2
REGIONAL LYMPH NODES: pN1a
Total number of lymph nodes examined: 5
Number of lymph nodes involved: 1
Number of sentinel nodes examined: 5
Number of sentinel nodes involved: 1

GrouStage 1B

Number of lymph nodes with macrometastases: 1
Number of lymph nodes with micrometastases: 0
Number of lymph nodes with isolated tumor cells: 0
Size of largest metastatic deposit: 2.1 mm
Extranodal extension: Not identified
DISTANT METASTASIS: Not applicable

BREAST PATHOLOGIC PROGNOSTIC GROUP MARKERS

OVERALL HISTOLOGIC GRADE (Nottingham histologic grade): Grade 2
ANCILLARY STUDIES: (Previously performed on prior biops[redacted]
HER 2/neu FISH: Negative/Not amplified
RATIO OF AVERAGE HER2/CEN-17: 1.1
AVERAGE NUMBER OF HER2 SIGNALS/NUCLEUS: 2.6
ESTROGEN RECEPTOR: Positive (90%)
PROGESTERONE RECEPTOR: Positive (60%)
MIB-1 (Ki-67): 15%

MICROSCOPIC:

TUMOR SITE: Left breast central and lower outer quadrant
TUMOR SIZE: Numerous varying sized foci of invasive carcinoma arising in close proximity to one another that coalesce together to form a single large confluent tumor mass spanning 39 mm in greatest dimension that is located within the lower outer/central breast specimen at approximately 4-5 o clock
HISTOLOGIC TYPE: Invasive ductal carcinoma, not otherwise specified
HISTOLOGIC GRADE:
Glandular (Acinar)/Tubular differentiation: Score 2
Nuclear pleomorphism: Score 2
Mitotic rate: Score 2
Nottingham histologic score: 6 of 9
Overall Grade: Grade 2
TUMOR FOCALITY: Single focus of invasive carcinoma
DUCTAL CARCINOMA IN-SITU (DCIS): Present
Size (extent) of DCIS: Associated with invasive carcinoma
Architectural Pattern(s): Micropapillary and papillary types
Nuclear grade: Grade 2 of 3
LOBULAR CARCINOMA IN SITU (LCIS): Not identified
TUMOR EXTENT:
Skin invasion: No skin is present
Skin satellite foci: Not applicable
Nipple: Nipple is not present
Skeletal muscle: No skeletal muscle is present
MARGINS:
Invasive carcinoma margins: Negative for invasive carcinoma
Distance from closest margin: A couple of fibroblasts between the inked margin and carcinoma
Specify margin: Posterior
Anterior margin: Less than 1 mm
Posterior margin: A couple of fibroblasts between the inked margin and carcinoma
Superior margin: Greater 10 mm
Inferior margin: Greater than 10 mm
Medial margin: Greater than 10 mm
Lateral margin: Greater than 10 mm

DCIS margins: Negative for Ductal carcinoma in situ
Distance from closest margin: Less than 1 mm
Specify margin: Anterior and Posterior margins
Anterior margin: Less than 1 mm
Posterior margin: Less than 1 mm
Superior margin: Greater than 10 mm
Inferior margin: Greater than 10 mm
Medial margin: Greater than 10 mm
Lateral margin: Greater than 10 mm
TREATMENT EFFECT:
Treatment effect in the breast: No known presurgical therapy
Treatment effect in the lymph nodes: No known presurgical therapy
LYMPHOVASCULAR INVASION: Not identified

MACROSCOPIC

PROCEDURE: Bilateral skin-sparing total mastectomies with left axillary sentinel lymph nodes #1, #2, and #3 with additional right nipple base and left nipple base excisions

Comment:

A) The right breast skin-sparing total mastectomy specimen demonstrates areas of flat epithelial atypia present within a background of prominent benign proliferative changes that includes adenosis, sclerosing adenosis, usual ductal hyperplasia, columnar cell hyperplasia, and stromal fibrosis. Within the upper/central inner quadrant is a 1.5-cm cyst containing dense inspissated secretions that likely represent the radiologic findings of a well-circumscribed oval homogeneous hypoechoic mass located at the 1-2 o'clock 1 cm from the nipple lesion. No evidence of ductal carcinoma in situ or invasive carcinoma is identified.

C) The left breast skin-sparing total mastectomy specimen demonstrates numerous varying sized invasive ductal carcinomas arising in close proximity to one another that coalesce together to form a single large confluent tumor mass located within the lower central/outer quadrant of the breast specimen at approximately 4-5 o clock spanning 39 mm in greatest dimension. Three (3) metal biopsy clips are associated the larger areas of the overall tumor mass. Prominent intermediate grade micropapillary and papillary-type DCIS is associated with the invasive carcinoma. Invasive carcinoma comes exceedingly close to the posterior margin with only a couple of fibroblasts separating the tumor from the inked margin. Invasive carcinoma also comes exceedingly close to the anterior margin at less than 1 mm. Ductal carcinoma in situ also comes very close to the anterior and posterior margin at less than 1 mm. All the other margins are negative at greater than 10 mm.
The sentinel lymph node #1 from the left axilla demonstrates metastatic carcinoma spanning 2.1 mm in greatest dimension. The remaining sentinel lymph nodes #2 and #3 are negative for metastatic carcinoma by routine H&E and pancytokeratin immunohistochemistry.
Overall, this is staged at pT2 pN1a.

Selected slides from this are submitted for intradepartmental consultation with satisfactory agreement.

*** Report Electronically Signed ***
psk/1/11/2022 George M. Sneed, D.O.

Gross Description

A) Received fresh from the OR for permanent section labeled "right breast, short = subareolar plexus, long = 12 o'clock" is a skin-sparing total mastectomy specimen. A long black suture demarcates the superior, 12 o'clock aspect of the specimen. A short black suture is present demarcating the subareolar plexus. The specimen measures 17.5 cm from superior to inferior, 15.0 cm from medial to lateral, and 1.8 cm from anterior to posterior. The specimen is inked as follows: The subareolar plexus is inked orange; the anterior is inked blue; and the posterior aspect is inked black. The specimen is then serially sectioned from lateral to medial into 9 slices. Examining the slices reveal abundant dense breast stroma seen throughout the breast tissue most predominant within the central breast tissue. Scattered

through the breast parenchyma are multiple small cysts ranging in size from 0.1 up to 0.3 cm in greatest dimension. Also seen within the upper mid to inner quadrant of the breast within slice 7 is a predominantly circumscribed nodular area within the dense breast stroma measuring 1.5 x 1.0 x 0.9 cm. No other distinct masses or lesions are identified. Representative sections from the specimen are submitted per the slide key.

SLIDE KEY:

A1-3) Slice 7, area of nodularity within the upper mid to inner quadrant, entirely submitted
A4) Slice 3, upper outer quadrant
A5) Slice 4, lower outer quadrant
A6-7) Subareolar plexus and retroareolar tissue
A8) Slice 8, lower inner quadrant
A9) Slice 8, upper inner quadrant

B) Received fresh from the OR for permanent section labeled "additional anterior margin at right nipple base, clips mark new margin" is a roughly rectangular fragment of pink-tan tissue that measures 1.2 x 1.2 x 0.3 cm. Two staples are present along one side demarcating the new true margin. The new true margin is inked blue, and the opposite margin is inked black. The specimen is then serially sectioned and entirely submitted in cassette B1.

C) Received fresh from the OR for radiologic assessment of clip placement labeled "left breast, short = subareolar plexus, long = 12 o'clock" is a skin-sparing total mastectomy specimen that measures 20.0 cm from superior to inferior, 15.5 cm from medial to lateral, and 1.8 cm from anterior to posterior. There is a short black suture along the anterior surface demarcating the nipple base. A long black suture is present along one edge demarcating the 12 o'clock superior aspect of the specimen. The specimen is inked as follows: The area demarcated as the nipple base is inked orange, the anterior surface is inked blue, and the posterior aspect is inked black. The specimen is then serially sectioned from medial to lateral into ten slices. Within slice 7 at approximately 4-5 o'clock are two adjacent HydroMARK biopsy clips associated with a dense white firm fibrous-type lesion. This lesion has vague borders and measures 3.9 x 1.6 x 1.0 cm. This lesion comes to within 1.0 cm from the anterior margin, 0.1 cm from the posterior margin, 10.0 cm from the superior margin, 3.0 cm from the inferior margin, 9.0 cm from the medial margin, and 5.0 cm from the lateral margin. Within slice 8 there is a lesion present that appears to be extending from the lesion seen within slice 7 that appears irregular, white-firm, and nodular measuring 1.8 x 1.1 x 1.6 cm and contains a non-HydroMARK metal biopsy clip. This lesion associated within slice 8 abuts the anterior and posterior margins, comes to within 8.5 from the superior margin, 1.1 from the inferior margin, 3.5 from the lateral margin, and 10.5 cm from the lateral margin. The tumor grossly is suspicious to extent in to slice 9 focally. The two lesions seen together within slices 7 and 8 have an overall measurement of 3.9 x 3.5 x 1.6 cm. The remaining breast tissue contain abundant dense stromal fibrosis and scattered small cysts ranging in size from 0.1 up to 0.3 cm in greatest dimension. No other gross masses or lesions are identified. Representative sections from the specimen are submitted per the slide key.

SLIDE KEY:

C1-16) Slice 7 containing the two biopsy clip area and lesion from superior to inferior with C1-4 consecutive sections from superior to inferior, C6-8 consecutive sections from superior to inferior, C9-12 consecutive sections from superior to inferior, and C13-16 consecutive sections from superior to inferior
C17-27) Slice 8 from superior to inferior containing adjacent/continuous lesion from slice 7 with C17-20 quadrisected, C21-24 quadrisected with C22 containing area of biopsy clip, C25-27 trisected with C26 containing the inferior margin
C28-29) Subareolar plexus from slice 4
C30) Slice 2, upper inner quadrant
C31) Slice 3, lower inner quadrant
C32) Slice 9, upper outer quadrant
C33) Slice 10, lower outer quadrant
C34) Slice 9, adjacent/lateral to lesion
C35-36) Slice 6, adjacent/medial to the lesion

D) Received fresh from the OR for permanent section labeled "additional anterior margin at left nipple base, clips mark new margin" is an irregularly-shaped fragment of pink-tan tissue that measures 0.9 x 0.6 x 0.2 cm. Two staples

are present along one side demarcating the true new margin. The true margin is inked blue, and the opposite margin is inked black. The specimen is then bisected and entirely submitted in cassette D1.

E) Received in formalin labeled "sentinel lymph node #1, left axilla, hot, #3541" is a single oval fragment of pink-tan tissue that measures 2.2 x 1.5 x 1.0 cm. This portion of tissue is searched for lymph nodes and one possible lymph node is identified measuring 1.8 x 1.3 x 1.0 cm. This possible lymph node is submitted per the sentinel lymph node protocol in cassettes E1-3.

F) Received in formalin labeled "sentinel lymph node #2, left axilla, hot, #1837" are multiple pink-yellow lobulated fragments of tissue with the smallest measuring 0.8 and the largest measuring 2.0 cm in greatest dimension. These fragments of tissue are searched for lymph nodes, and three possible lymph nodes are identified ranging in size from 0.6 up to 1.3 cm in greatest dimension. These possible lymph nodes are submitted per the sentinel lymph node protocol per the slide key.

SLIDE KEY:

F1) One possible lymph node, serially sectioned
F2) One possible lymph node, serially sectioned
F3) One possible lymph node, serially sectioned

G) Received in formalin labeled "sentinel lymph node #3, left axilla, hot, #948" is a single oval fragment of yellow-pink adipose tissue that measures 2.5 x 1.4 x 0.7 cm. This portion of tissue is searched for lymph nodes, and one possible lymph node is identified measuring 1.6 x 1.1 x 0.3 cm. This possible lymph node is serially sectioned per the sentinel lymph node protocol and is entirely submitted in cassettes G1-2.

GMS/psk/1/8/2022

Microscopic Description

C) Documentation of Medical Necessity for Immunohistochemistry:

Immunostains for a breast cocktail containing p63 and heavy chain smooth muscle myosin is performed for further assessment of the extent of the invasive carcinoma and ductal carcinoma in situ. A pancytokeratin immunostain is also performed for further assessment of the presence of tumor cells in relation to the posterior margin. An E cadherin immunostain is also performed for further assessment of ductal carcinoma, lobular in origin.

Description of Immunohistochemistry:

A breast cocktail immunostain containing p63 and heavy chain smooth muscle myosin is performed with adequate controls on blocks C11, C17, C18, C20, C22, C31, and C34. The breast cocktail immunostained sections highlight a loss of myoepithelial cells within the areas of interest, supportive of invasive carcinoma. A pancytokeratin immunostain is performed with adequate controls on block C18. The pancytokeratin immunostained section highlights the area of invasive carcinoma in relation to the margins of excision. An E cadherin immunostain is performed with adequate controls on block C25 and demonstrates diffuse staining, supportive of ductal in origin.

Immunostain Summary:
C11: Breast cocktail containing p63 and heavy chain smooth muscle myosin
C17: Breast cocktail containing p63 and heavy chain smooth muscle myosin
C18: Breast cocktail containing p63 and heavy chain smooth muscle myosin, Pancytokeratin
C20: Breast cocktail containing p63 and heavy chain smooth muscle myosin
C22: Breast cocktail containing p63 and heavy chain smooth muscle myosin
C25: E cadherin
C31: Breast cocktail containing p63 and heavy chain smooth muscle myosin
C34: Breast cocktail containing p63 and heavy chain smooth muscle myosin

E) Documentation of Medical Necessity for Immunohistochemistry:

Per protocol given the lymph node is a sentinel lymph node, a pancytokeratin immunostain is performed for further assessment of the presence of isolated tumor cells and micrometastasis.

Description of Immunohistochemistry:

A pancytokeratin immunostain is performed with adequate controls on blocks E1, E2, and E3. The pancytokeratin immunostained sections highlight small foci of metastatic carcinoma within section E1.

Immunostain Summary:
E1: Pancytokeratin
E2: Pancytokeratin
E3: Pancytokeratin

F) Documentation of Medical Necessity for Immunohistochemistry:

Per protocol given the lymph node is a sentinel lymph node, a pancytokeratin immunostain is performed for further assessment of the presence of isolated tumor cells and micrometastasis.

Description of Immunohistochemistry:

A pancytokeratin immunostain is performed with adequate controls on blocks F1, F2, and F3. The pancytokeratin immunostained sections are negative for metastatic carcinoma.

Immunostain Summary:
F1: Pancytokeratin
F2: Pancytokeratin
F3: Pancytokeratin

G) Documentation of Medical Necessity for Immunohistochemistry:

Per protocol given the lymph node is a sentinel lymph node, a pancytokeratin immunostain is performed for further assessment of the presence of isolated tumor cells and micrometastasis.

Description of Immunohistochemistry:

A pancytokeratin immunostain is performed with adequate controls on blocks G1 and G2. The pancytokeratin immunostained sections are negative for metastatic carcinoma.

Immunostain Summary:
G1: Pancytokeratin
G2: Pancytokeratin

In some tests, analyte specific reagents (ASRs) are used. In the case of an ASR, this test was developed and its performance characteristics determined by this laboratory. It has not been cleared or approved by the US Food and Drug Administration. The FDA has determined that such clearance or approval is not necessary. This test is used for clinical purposes. It should not be regarded as investigational or for research. This laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA '88) as qualified to perform high complexity clinical laboratory testing.

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Knoxville, TN 37920

Surgical Pathology Report

Patient: [redacted]
Med. Rec. #:
DOB/Gender:
Soc. Sec. #:
Location:
Physician(s):

[redacted]

Specimen(s) Received

A:Level 9 lymph node
B:Level 7 lymph node
C:Level 10 lymph node
D:Level 11 lymph node
E:Level 11 lymph node #2
F:Level 10 lymph node #2
G:Level 11 lymph node #3
H:Left upper lobe
I:Level 5 lymph node

Clinical History and Diagnosis

Squamous cell carcinoma of left upper lobe of lung

Final Pathologic Diagnosis

A) **LYMPH NODE, LEVEL 9, EXCISION :**
NEGATIVE FOR MALIGNANCY IN TWO LYMPH NODES (0/2)

B) **LYMPH NODE, LEVEL 7, EXCISION:**
NEGATIVE FOR MALIGNANCY IN TWO LYMPH NODES (0/2)

C) **LYMPH NODE, LEVEL 10, EXCISION:**
NEGATIVE FOR MALIGNANCY IN ONE LYMPH NODE (0/1)

D) **LYMPH NODE, LEVEL 11, EXCISION:**
NEGATIVE FOR MALIGNANCY IN TWO LYMPH NODES (0/2)

E) **LYMPH NODE, LEVEL 11, NODE #2, EXCISION:**
NEGATIVE FOR MALIGNANCY IN ONE LYMPH NODE (0/1)

F) **LYMPH NODE, LEVEL 10, NODE #2, EXCISION:**
NEGATIVE FOR MALIGNANCY IN THREE LYMPH NODES (0/3)

G) LYMPH NODE, LEVEL 11, NODE #3, EXCISION:
NEGATIVE FOR MALIGNANCY IN FOUR LYMPH NODES (0/4)

H) LUNG, LEFT UPPER LOBE, LOBECTOMY:
INVASIVE SQUAMOUS CELL CARCINOMA, NON-KERATINIZING
TUMOR SIZE: 1.6 CM
HISTOLOGIC GRADE: POORLY-DIFFERENTIATED (G3)
MARGINS: NEGATIVE FOR MALIGNANCY
DISTANCE TO CLOSEST MARGIN: 3.5 CM FROM BRONCHIAL / VASCULAR MARGIN
VISCERAL PLEURAL INVASION: NOT IDENTIFIED
ANGIOLYMPHATIC INVASION: PRESENT
TUMOR IDENTIFIED WITH DIRECT EXTENSION INTO WALL OF LARGE CALIBER BLOOD VESSEL
PERINEURAL INVASION: NOT IDENTIFIED
NON-NEOPLASTIC LUNG: EMPHYSEMATOUS CHANGES

PERIBRONCHIAL LYMPH NODES: POSITIVE FOR MALIGNANCY IN ONE OF ELEVEN LYMPH NODES BY DIRECT TUMOR EXTENSION (1/11), SEE COMMENT

I) LYMPH NODE, LEVEL 5, EXCISION:
NEGATIVE FOR MALIGNANCY IN ONE LYMPH NODE (0/1)

SYNOPTIC SUMMARY

PATHOLOGIC STAGING
PRIMARY TUMOR: pT1b
REGIONAL LYMPH NODES: pN1
Number of lymph nodes with tumor: 1
Nodal site(s) with tumor: Peribronchial node with involvement by direct tumor extension
Number of lymph nodes examined: 27
Nodal site(s) examined: Peribronchial and levels 5, 7, 9, 10, 11
Lymph nodes from prior procedures: Not applicable
DISTANT METASTASES: Not applicable

Group Stage 2B

TUMOR SIZE
TOTAL TUMOR SIZE:
Greatest dimension: 1.6 CM
Additional dimensions: 1.1 x 1.1 CM
INVASIVE TUMOR SIZE:
Greatest dimension: 1.6 CM
Additional dimensions: 1.1 x 1.1 CM

MICROSCOPIC
HISTOLOGIC TYPE: Invasive squamous cell carcinoma, non-keratinizing
Histologic components present: Squamous cell carcinoma
Histologic patterns present: Squamous cell carcinoma, non-keratinizing
VISCERAL PLEURA INVASION: Not identified
LYMPH-VASCULAR INVASION: Present
DIRECT INVASION OF ADJACENT STRUCTURES: Not applicable
Involved adjacent structures: Not applicable
MARGINS:

Margin status for invasive carcinoma: All margins negative for invasive carcinoma
Closest margin(s) to invasive carcinoma: Bronchial / vascular
Distance from invasive carcinoma to closest margin: 3.5 CM
Margins involved by invasive carcinoma: Not applicable
Margin status for non-invasive tumor: Not applicable
TREATMENT EFFECT: No known presurgical therapy
Percentage of residual viable tumor: Not applicable

MACROSCOPIC
PROCEDURE: Lobectomy
SPECIMEN LATERALITY: Left
TUMOR SITE: Upper lobe
TUMOR SIZE: 1.6 CM
TUMOR FOCALITY: Unifocal
SYNCHRONOUS TUMORS: Not applicable

Comment:

There is one large peribronchial lymph node in immediate region of tumor showing involvement of squamous cell carcinoma by likely direct tumor extension.

Immunostains performed on prior biopsy [redacted] support tumor to be of squamous cell origin.
One portion of fresh tissue from tumor [redacted] is provided to UTMC Biobank.

Tissue blocks with viable tumor for additional studies, if requested: H5, H6

*** Report Electronically Signed ***
ctc/3/10/2022 Christopher T. Clark, MD

Addendum

Addendum Diagnosis

REASON FOR ADDENDUM: Additional information added without change in diagnosis
DATE OF ADDENDUM: 05/20/2022
ADDENDUM NUMBER: 1

Additional testing performed by Integrated Oncology, Brentwood, TN ([redacted]), reporting the following on 05/20/2022:

Test Performed: EGFR Gene Mutation Analysis

Result: Negative for EGFR mutation

This is only partial information from the outside report. Copy of full outside report with additional information is available in the Department of Pathology upon request, or is available in PowerChart under "outside pathology reports".

*** Report Electronically Signed ***
Christopher T. Clark, MD

Gross Description

A) Received in formalin labeled with the patient's name, [redacted] and "level 9 lymph node" are two brown-yellow fragments of soft tissue. The larger fragment measures 0.8 cm and the smaller 0.4 cm in greatest dimension. The entire specimen is submitted as A1.

B) Received in formalin labeled with the patient's name, [redacted] and "level 7 lymph node" are two anthracotic possible lymph nodes. The larger measures 1.7 x 1.4 x 0.6 cm and the smaller measures 1.3 x 1 x 0.5 cm. Both are bisected and the larger is entirely submitted as B1 and the smaller entirely submitted as B2.

C) Received in formalin labeled with the patient's name, [redacted] and "level 10 lymph node" is a black-yellow possible lymph node that measures 3 x 1.5 x 1 cm. The specimen is bisected and is entirely submitted as C1-C2.

D) Received in formalin labeled with the patient's name, [redacted] and "level 11 lymph node" are four black-brown possible lymph nodes that range in size from 0.1 cm up to 0.5 cm in greatest dimension. The entire specimen is submitted as D1.

E) Received in formalin labeled with the patient's name, [redacted] and "level 11 lymph node #2" is a black-brown possible lymph node that measures 1 x 0.8 x 0.5 cm. The specimen is bisected and entirely submitted in E1.

F) Received in formalin labeled with the patient's name, [redacted] and "level 10 lymph node #2" are three anthracotic possible lymph nodes ranging in size from 0.3 cm up to 0.8 cm in greatest dimension. The entire specimen is submitted as F1.

G) Received in formalin labeled with the patient's name, [redacted] and "level 11 lymph #3" are four black possible lymph nodes ranging in size from 0.2 cm up to 2.5 cm in greatest dimension. The largest is bisected and entirely submitted as G1-G2 and the three smaller entirely as G3.

(cassettes verified)

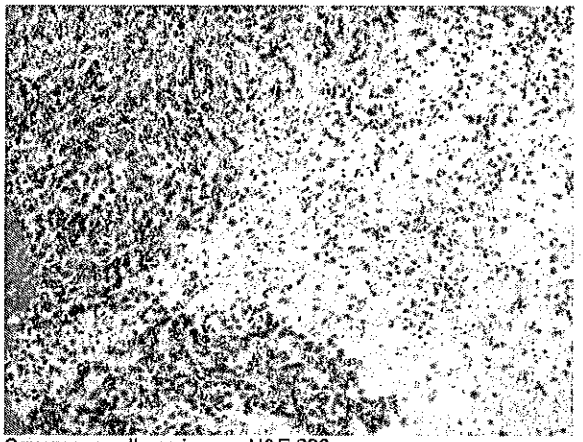
H) Received fresh for intraoperative frozen section diagnosis labeled with the patient's name, [redacted] and as "left upper lobe" is a lobe of lung weighing 235 grams and measuring 20 x 10.5 x 2.5 cm. The pleural surface is black-tan and smooth throughout. There is a single firm, white pleural nodule measuring 0.4 cm in diameter. This nodule is located 9 cm from the bronchial/vascular staple line margin. The vascular and bronchial margins are stapled. The staple line is removed and the bronchial margin is submitted in cassette H1 for intraoperative frozen section diagnosis. Located 2.5 cm from the firm pleural nodule and 3.5 cm from the bronchial/vascular margin is a firm, circumscribed, white nodule measuring 1.6 x 1.1 x 1.1 cm. This lesion is located 1.5 cm beneath the nearest pleural surface. The overlying pleural surface is smooth with no puckering evident. One representative piece of mass is submitted to UTMC Biobank. The remaining lung tissue is tan-brown with no other nodules or masses identified. No additional staple line at the parenchymal surface is noted. Representative sections including the entire mass are submitted with processing in formalin under the following designations:

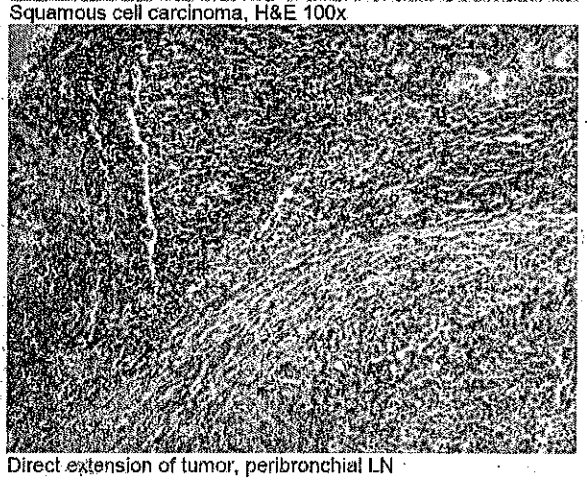
SLIDE KEY:

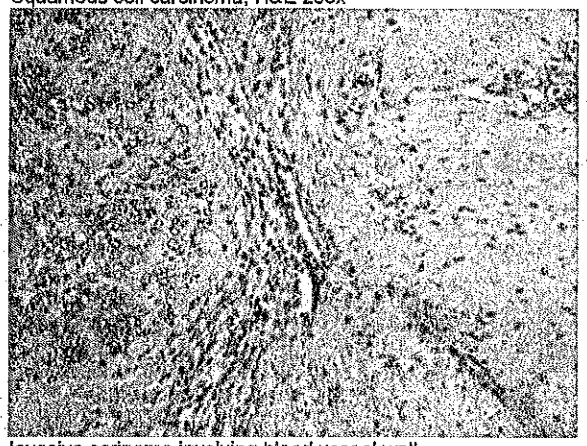
H1FS)	Shaved bronchial margin
H2)	Firm pleural nodule after decalcification
H3)	Representative tissue between firm pleural nodule and mass
H4)	Pleural surface closest to mass
H5-H10)	Mass, submitted entirely
H11)	Shaved vascular margin
H12-H13)	Multiple peribronchial lymph nodes
H14)	Single lymph node, bisected
H15)	Single lymph node, bisected
H16-H17)	Single lymph node, bisected
H18-H19)	Representative uninvolved lung parenchyma

I) Received in formalin labeled with the patient's name, [redacted] and "level 5 lymph node" is a single piece of tan tissue measuring 1.2 x 0.3 x 0.1 cm. Embedded in toto in cassette I1.


Squamous cell carcinoma, H&E 100x


Squamous cell carcinoma, H&E 200x


Direct extension of tumor, peribronchial LN


Invasive carinoma involving blood vessel wall

CTC/rjc/3/10/2022

Intraoperative Consult Diagnosis

H) Left upper lobe lung:
(H1FS): Shaved bronchial margin:
Negative for malignancy

Christopher T. Clark, M.D.
Pager: 3954
Date: 03/10/2022
Time: 1211

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Surgical Pathology Report

Patient: [redacted]
Med. Rec. #:
DOB/Gender:
Soc. Sec. #:
Location:
Physician(s):

[redacted]

Specimen(s) Received

A:Right pelvic lymph node
B:Left pelvic lymph node
C:Prostate

Clinical History and Diagnosis

Prostate cancer (per PowerChart), C61

Final Pathologic Diagnosis

A) LYMPH NODES, RIGHT PELVIC, LYMPH NODE DISSECTION:
NEGATIVE FOR MALIGNANCY IN TEN LYMPH NODES (0/10)

B) LYMPH NODES, LEFT PELVIC, LYMPH NODE DISSECTION:
NEGATIVE FOR MALIGNANCY IN SIX LYMPH NODES (0/6)

C) PROSTATE, RADICAL PROSTATECTOMY:
ACINAR ADENOCARCINOMA
GLEASON PATTERN/SCORE: 4+5 = 9
GRADE GROUP: 5
EXTENT: INVOLVES APPROXIMATELY 60-70% OF GLAND
EXTRAPROSTATIC EXTENSION: PRESENT
SEMINAL VESICLE INVASION: PRESENT
MARGINS: NEGATIVE FOR MALIGNANCY
ANGIOLYMPHATIC INVASION: NOT IDENTIFIED
PERINEURAL INVASION: PRESENT

SYNOPTIC SUMMARY

PATHOLOGIC STAGING
PRIMARY TUMOR: pT3b
REGIONAL LYMPH NODES: pN0
Number of lymph nodes with tumor: 0

Group Stage 3B

Number of lymph nodes examined: 16
DISTANT METASTASES: Not applicable

MICROSCOPIC
HISTOLOGIC TYPE: Acinar adenocarcinoma
HISTOLOGIC GRADE: Grade group 5 (Gleason Score 4+5=9)
Percentage of pattern 4: 60-70%
Intraductal carcinoma (IDC): Not identified
IDC incorporated into grade: Not applicable
Cribriform glands: Present
TUMOR QUANTITATION:
Estimated percentage of prostate involved by tumor: 60%
Greatest dimension of dominant nodule: 45 mm
EXTRAPROSTATIC EXTENSION: Present
URINARY BLADDER NECK INVASION: Not identified
SEMINAL VESICLE INVASION: Present
LYMPHOVASCULAR INVASION: Not identified
MARGIN STATUS: All margins negative for invasive carcinoma
Margin(s) involved by invasive carcinoma: Not applicable
TREATMENT EFFECT: No known presurgical therapy
ADDITIONAL FINDINGS: None identified

MACROSCOPIC
PROCEDURE: Radical prostatectomy
PROSTATE SIZE:
Weight: 51 grams
Size: 5.0 X 4.5 X 4.3 CM

ctc/2/1/2022

*** Report Electronically Signed ***
Christopher T. Clark, MD

Gross Description

A) Received in formalin, labeled with the patient's name, [redacted], and "right pelvic lymph node" consists of an 8.2 x 3 x 2.4 cm aggregate of yellow, lobulated, adipose tissue admixed with lymphatic tissue. There are 11 tan-pink, fatty replaced, potential lymph nodes identified, ranging from 0.2 cm to 2 cm. The lymphatic tissue entirely submitted as follows:

SLIDE KEY:
A1) Three whole potential lymph nodes
A2) Two whole potential lymph nodes
A3) One whole potential lymph node
A4) One potential lymph node, bisected
A5) One potential lymph node, bisected
A6-A7) One potential lymph node, bisected
A8-A9) One potential lymph node, bisected
A10-A11) One potential lymph node, serially sectioned

B) Received in formalin, labeled with the patient's name, [redacted] and "left pelvic lymph node" consists of a 5.8 x 3 x 1.4 cm aggregate of yellow, lobulated adipose tissue admixed with lymphatic tissue. There are seven tan-pink, fatty-replaced potential lymph nodes identified, ranging from 0.3 cm to 1.9 cm. The lymphatic tissue is submitted entirely as follows:

SLIDE KEY:
B1) Three whole potential lymph nodes

B2) Two whole potential lymph nodes
B3) One potential lymph node, bisected
B4-B5) One potential lymph node, bisected

C) Received in formalin, labeled with the patient's name, [redacted], and "prostate" consists of a 4.3 cm apex to base, 5 cm left to right, and 4.5 cm anterior to posterior prostate gland with attached bilateral adnexa. There is a 2.5 x 1.5 x 0.5 cm left seminal vesicle that is fragmented and loosely attached, a 1.4 cm in length x 0.5 cm in diameter left vas deferens, a 3 x 2.3 x 0.9 cm right seminal vesicle, and a 1.7 cm in length x 0.5 cm in diameter right vas deferens. The capsule is tan-brown and diffusely cauterized and appears intact. The prostatic urethra is probe patent. The prostate gland devoid of adnexa is 51 grams. The left anterior and lateral half is inked blue, the right anterior and lateral half is inked orange, and the posterior is inked black. The specimen is serially sectioned from apex to base into seven slices to reveal pale tan-pink, diffusely nodular parenchyma. Representative sections are submitted as follows:

SLIDE KEY:
C1) Longitudinal section of left seminal vesicle and cross-section of left vas deferens
C2) Longitudinal section of right seminal vesicle and cross-section of right vas deferens
C3) Section of junction of left seminal vesicle, left vas deferens, and prostate
C4) Junction of right seminal vesicle, right vas deferens, and prostate
C5) Right angle sections of left apex
C6) Right angle sections of right apex
C7) Right angle sections of the left bladder base
C8) Right angle sections of the right bladder base
C9) Slice 1-left anterior
C10) Slice 1-left posterior
C11) Slice 1-right anterior
C12) Slice 1-right posterior
C13) Slice 2-left posterior
C14) Slice 2-right posterior
C15) Slice 3-left anterior
C16) Slice 3-left posterior
C17) Slice 3-right anterior
C18) Slice 3-right posterior
C19-C20) Slice 4-left posterior
C21-C22) Slice 4-right posterior
C23) Slice 5-left anterior
C24-C25) Slice 5-left posterior
C26) Slice 5-right anterior
C27-C28) Slice 5-right posterior
C29) Slice 6-left posterior
C30) Slice 6-right posterior
C31) Slice 7-left anterior
C32) Slice 7-left posterior
C33) Slice 7-right anterior
C34) Slice 7-right posterior

(cassettes verified)

TMS/rjc/1/31/2022

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Surgical Pathology Report

[redacted]

Location: [redacted]
Physician(s):

Specimen(s) Received

A:Right renal mass

Clinical History and Diagnosis

Right renal mass
Operation: Robotic right partial nephrectomy

Final Pathologic Diagnosis

A) RIGHT RENAL MASS, ROBOTIC RIGHT PARTIAL NEPHRECTOMY:
PAPILLARY RENAL CELL CARCINOMA, 5.6 cm IN GREATEST DIMENSION, FUHRMAN NUCLEAR GRADE 1
OUTER SURFACES OF THE TUMOR ARE DISRUPTED HAMPERING ACCURATE EVALUATION OF MARGINS; WHERE INTACT, TUMOR APPEARS CONFINED TO THE KIDNEY.
NO NECROSIS OR SARCOMATOID FEATURES ARE IDENTIFIED.
PATHOLOGIC STAGE: pT1b pN-NOT ASSIGNED (NO LYMPH NODES SUBMITTED OR FOUND)
SEE SYNOPTIC SUMMARY BELOW

SYNOPTIC SUMMARY

PATHOLOGIC STAGE CLASSIFICATION (pTNM, AJCC 8th Edition):
TNM descriptors: Not applicable
PRIMARY TUMOR (pT): pT1b
REGIONAL LYMPH NODES (pN): pN – not assigned (no nodes submitted or found)
DISTANT METASTASIS (pM): Not applicable – pM cannot be determined from the submitted specimen(s)

Unable to group stage, no lymph nodes removed

MICROSCOPIC
HISTOLOGIC TYPE: Papillary renal cell carcinoma
SARCOMATOID FEATURES: Not identified
RHABDOID FEATURES: Not identified
HISTOLOGIC GRADE (WHO / ISUP Grade): G1

[redacted]

TUMOR NECROSIS: Not identified
TUMOR EXTENSION: Limited to kidney where not disrupted
MARGINS: Cannot be determined: All margins negative for invasive carcinoma where not disrupted
REGIONAL LYMPH NODE STATUS: Not applicable (no regional lymph nodes submitted or found)
ADDITIONAL FINDINGS IN NONNEOPLASTIC KIDNEY: Insufficient tissue

MACROSCOPIC
PROCEDURE: Partial nephrectomy
SPECIMEN LATERALITY: Right
TUMOR SIZE: Greatest dimension: 5.6 cm
TUMOR FOCALITY: Unifocal

Comment:

Selected slides are seen in intradepartmental consultation with satisfactory agreement.

*** Report Electronically Signed ***
psk/1/13/2022 Amila Orucevic, M.D., Ph.D.

Gross Description

A) Received in formalin labeled [redacted], with date of birth, and "right renal mass" is a multifocally disrupted, non-intact irregularly shaped tumor mass which consists of a multifocally disrupted thin and delicate capsule. Where capsule is intact, it is inked blue. The capsule is associated with a soft to semisolid tumor that measures 5.6 x 5.5 x 3.8 cm and which, also, is multifocally disrupted. There is no grossly apparent renal parenchyma. Focally, on the exterior surface of the thin capsule is lobulated adipose tissue that appears unremarkable. The cut surfaces of the tumor tissue are uniformly solid and demonstrate significant areas of hemorrhage and softening. The cut surfaces demonstrate tiny rim of possible renal parenchyma in some areas which are not disrupted. Representative sections are submitted as follows:
A1-A6) Representative sections of tumor, taken from areas with distinct gross appearance
A7) Representative adipose tissue and membranous tissue away from tumor

(cassettes verified)

ELH/psp/1/11/2022

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Knoxville, TN 37920

Surgical Pathology Report

Patient: [redacted]
Med. Rec. #:
DOB/Gender:
Soc. Sec. #:
Location:
Physician(s):

Specimen(s) Received

A:Left inguinal space sentinal lymph node #1
B:Palpable left inguinal space sentinal lymph node #2
C:WLE left lower thigh

Clinical History and Diagnosis

Melanoma

Final Pathologic Diagnosis

A) LYMPH NODE, "LEFT INGUINAL SPACE SENTINEL #1," EXCISION:
NEGATIVE FOR MALIGNANCY IN ONE LYMPH NODE

B) LYMPH NODE, "PALPABLE LEFT INGUINAL SPACE SENTINEL #2," EXCISION:
NEGATIVE FOR MALIGNANCY IN ONE LYMPH NODE

C) SKIN AND SUBCUTANEOUS TISSUE, "LEFT LOWER THIGH," WIDE LOCAL EXCISION:
1) NEGATIVE FOR RESIDUAL MELANOMA
2) BIOPSY SITE CHANGES
3) UNREMARKABLE MARGINS OF EXCISION

Group Stage 2B

SEE COMMENT

Comment:

As per information from the original biops[redacted], the final pathologic stage is pT4a N0.

psk/1/12/2022

*** Report Electronically Signed ***
Alan Grindstaff, M.D.

Gross Description

[redacted]

A) Received in formalin labeled [redacted] with date of birth, and "left inguinal space sentinel lymph node #1, 1827, hot, blue" is a focally cauterized 2.5 x 1.3 x 1.3 cm faintly blue-stained lymph node which is sectioned and entirely submitted according to Sentinel Lymph Node Protocol in A1-A4. (cassettes verified)

B) Received in formalin labeled [redacted] with date of birth, and "palpable left inguinal sentinel lymph node #2, hot, not blue, 100" is a 2.2 x 1.3 x 1 cm lymph node which is sectioned and entirely submitted according to Sentinel Lymph Node Protocol in B1-B4. (cassettes verified)

C) Received in formalin labeled [redacted] with date of birth, and "WLE left lower thigh, suture 12 o'clock superior" is an elliptical excision of thin, wrinkled, tan-white skin with subcutaneous tissue that has a suture at one tip marking the 12 o'clock. Using this orientation, the specimen measures 13.0 cm 12 to 6 o'clock x 4.0 cm 9 to 3 o'clock x 3.5 cm in thickness. The 12 to 3 to 6 o'clock is inked orange, the 6 to 9 to 12 o'clock is inked green, and the deep margin is inked black. On the skin surface is dark blue dye staining of the central portion, within which is a focally retracted and white-tan consistent with a biopsy site and focally dark brown and pigmented, with the overall lesional area measuring 1.0 cm from 12 to 6 o'clock x 0.7 cm from 9 to 3 o'clock and with this lesion located 1.5 cm from the 9 o'clock, 1.8 cm from the 3 o'clock, 6.2 cm from the 12 o'clock, and 5.8 cm from the 6 o'clock. On cut section is marked dark blue dye staining but no definite invasion of the subcutaneous tissue by the lesion. No other gross lesions are identified. Representative sections are submitted in cassettes as follows:

SLIDE KEY::

C1) 12 o'clock tip, en face
C2-C3) Bisected superficial half of slice, area before lesion
C4-C5) Bisected superficial half of slice with lesion
C6-C7) Bisected superficial half of slice with lesion
C8) Deep margin closest to lesion
C9-C10) Bisected superficial half of slice with lesion
C11-C12) Bisected superficial half of slice with lesion
C13-C14) Bisected superficial half of slice, area after lesion
C15) 6 o'clock tip, en face

(cassettes verified)

ELH/psp/1/4/2022

Microscopic Description

Immunostains for melanA and SOX10 have been reviewed on all sentinel lymph node blocks.

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Surgical Pathology Report

Patient: [redacted]
Med. Rec. #:
DOB/Gender:
Soc. Sec. #:
Location:
Physician(s):

Specimen(s) Received

A:Ileum, appendix, right colon, proximal transverse colon

Clinical History and Diagnosis

Cecal intramucosal adenocarcinoma s/p partial polypectomy

Final Pathologic Diagnosis

SMALL BOWEL AND COLON, RIGHT HEMICOLECTOMY:

1) ADENOCARCINOMA
   HISTOLOGIC GRADE: WELL-TO-MODERATELY-DIFFERENTIATED (GRADE 1-2)
   SIZE: 0.7 CM GREATEST DIMENSION
   MAXIMUM TUMOR EXTENSION: SUBMUCOSA
   MARGINS: NEGATIVE
   LYMPHOVASCULAR INVASION: NOT IDENTIFIED
2) BENIGN APPENDIX WITH DISTAL FIBROUS LUMINAL OBLITERATION
3) THIRTY-ONE LYMPH NODES, NEGATIVE FOR METASTATIC ADENOCARCINOMA (0/31)
4) SEE SYNOPTIC SUMMARY

SYNOPTIC SUMMARY

PATHOLOGIC STAGING:

PRIMARY TUMOR:	pT1
REGIONAL LYMPH NODES:	pN0
DISTANT METASTASIS:	Not applicable

Group Stage 1

ANCILLARY STUDIES:

IMMUNOHISTOCHEMISTRY TESTING FOR MISMATCH REPAIR PROTEINS:

Interpretation:	No loss of nuclear expression of MMR proteins: Low probability of MSI-H
Results:	
MLH1:	Intact nuclear expression, tumor cells
MSH2:	Intact nuclear expression, tumor cells
MSH6:	Intact nuclear expression, tumor cells

PMS2: Intact nuclear expression, tumor cells

MICROSCOPIC:

HISTOLOGIC TYPE: Adenocarcinoma
HISTOLOGIC GRADE: G1-2 (well-to-moderately-differentiated)
TUMOR EXTENT: Invades submucosa
MARGINS:
Margin status for invasive carcinoma: All margins negative for invasive carcinoma
Margin status for noninvasive tumor: All margins negative for high-grade dysplasia / intramucosal carcinoma, and low-grade dysplasia
TREATMENT EFFECT: No known presurgical therapy
LYMPHOVASCULAR INVASION: Not identified
PERINEURAL INVASION: Not identified
TYPE OF POLYP IN WHICH INVASIVE CARCINOMA AROSE: Tubular adenoma
REGIONAL LYMPH NODE STATUS: All regional lymph nodes negative for tumor
TUMOR DEPOSITS: Not identified

MACROSCOPIC:

PROCEDURE: Right hemicolectomy
TUMOR SITE: Cecum
TUMOR SIZE:
Greatest dimension: 0.7 cm
MACROSCOPIC TUMOR PERFORATION: Not identified

rjc/2/16/2022

*** Report Electronically Signed ***
Yorke D. Young, M.D.

Gross Description

Received fresh for Biobank, labeled with the patient's name [redacted] and "ileum, appendix, right colon, proximal transverse colon" consists of an anatomically oriented 7.2 cm in length x 2.0 cm in diameter portion of terminal ileum (2.5 cm of mesentery) and a 25.0 cm in length x 4.0 cm in diameter portion of right colon (8.0 cm of mesentery). The specimen has two stapled margins and there is a 6.2 cm in length x 0.7 cm in diameter unremarkable vermiform appendix grossly identified. The serosa is tan-pink and smooth with no visible perforations. The lumen is filled with a moderate amount of green viscous fecal debris and there is a wall thickness of 0.2 cm. The mucosa is tan-pink with multiple dark red petechial hemorrhages. There is a 0.7 x 0.6 cm puckered area (previous polypectomy site) in the area of the cecum that is 3.5 cm from the appendiceal orifice, 4.5 cm from the ileus valve, 13.0 cm from the proximal terminal ileum margin, and 24.0 cm from the distal colonic margin. The puckered area is located on the antimesenteric surface and the serosa overlying the area is inked black. Sectioning reveals a tan-white scarred cut surfaces that appear to be confined to the mucosa and submucosa and does not appear to involved the muscular layer. The remaining mucosa is unremarkable and in the terminal ileum there is a 1.5 x 1.0 cm tan-white discolored area on the serosa that is at the proximal resection margin and the overlying serosa is inked blue. The cut sections display homogenous white cut surfaces within the muscular wall. No other lesions or polyps are grossly identified. There are forty-five tan-pink possible lymph nodes grossly identified ranging from 0.2 to 1.0 cm. Representative sections are submitted as follows:

SLIDE KEY:

A1-A2) Proximal resection margin, perpendicular with discolored possible lesion (margin inked orange, overlying serosa inked blue)
A3) Distal colonic margin, en face
A4) Mesenteric margin, en face
A5-A7) Puckered area in cecum, entirely and sequentially (serosa inked black)

A8) Ileocecal valve
A9) Appendix
A10) Six whole possible lymph nodes
A11) Six whole possible lymph nodes
A12) Six whole possible lymph nodes
A13) Five whole possible lymph nodes
A14) Four whole possible lymph nodes
A15) Four whole possible lymph nodes
A16) Four whole possible lymph nodes
A17) Four whole possible lymph nodes
A18) Two possible lymph nodes, each bisected (one inked blue)
A19) Two possible lymph nodes, each bisected (one inked blue)
A20) One possible lymph node, bisected

(cassettes verified)

TMS/psp/2/16/2022

Microscopic Description

Documentation of Medical Necessity for Immunohistochemistry:

Evaluation for mismatch repair proteins as well as single, slightly enlarged lymph node with architectural effacement by relatively monotonous, moderate size lymphocyte population with coarse chromatin features

Description of Immunohistochemistry:

Immunostains are performed with adequate controls. There is intact nuclear expression of MLH1, MSH2, MSH6, and PMS2 within the tumor cells. Within the lymph node in question, CD3 and CD20 highlight T-cells and B-cells, respectively, with B-cells forming small nodular aggregates. There is paracortical expansion of T cells in the lymph node in question. CD5 and CD43 highlight T-cells, without aberrant expression on B-cells. CD23 highlights follicular dendritic cell meshworks associated with follicles. The germinal centers of secondary follicles coexpress CD10 and BCL6 without aberrant expression of BCL2, which highlights all T-cells and non-germinal center B-cells. Ki-67 is increased within the germinal centers, as expected, but not significantly increased otherwise. Cyclin D1 is negative. These finding support the diagnosis.

Immunostain Summary:
A6: MLH1, MSH2, MSH6, PMS2
A13: CD3, CD20, CD5, CD23, CD10, BCL2, Ki-67, CD43, Cyclin D1

In some tests, analyte specific reagents (ASRs) are used. In the case of an ASR, this test was developed and its performance characteristics determined by this laboratory. It has not been cleared or approved by the US Food and Drug Administration. The FDA has determined that such clearance or approval is not necessary. This test is used for clinical purposes. It should not be regarded as investigational or for research. This laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA '88) as qualified to perform high complexity clinical laboratory testing.

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Knoxville, TN 37920

Bone Marrow Report

[redacted] [redacted]

Specimen(s) Received

A:Bone marrow core biopsy
B:Bone marrow aspirate clot section
C:Fourteen bone marrow aspirate smears
D:Five peripheral blood smears

Clinical History and Diagnosis

Mantel cell lymphoma

Final Pathologic Diagnosis

BONE MARROW; CORE BIOPSY, CLOT SECTION, ASPIRATE SMEARS AND FLOW CYTOMETRY:
1) SUSPICIOUS FOR LOW LEVEL INVOLVEMENT BY MANTEL CELL LYMPHOMA
2) NORMOCELLULAR BONE MARROW FOR AGE (30%) WITH TRILINEAGE HEMATOPOIESIS WITH ADEQUATE MATURATION
3) SEE COMMENT AND MICROSCOPIC DESCRIPTION

PERIPHERAL BLOOD, SMEAR REVIEW:
1) NORMOCYTIC ANEMIA (HGB - 10.1 g/dL, MCV - 81.9 fL) - RETICULOCYTE PRODUCTION INDEX: 0.49
2) SEE COMMENT AND MICROSCOPIC DESCRIPTION

NO TNM on Lymphoma
Group Stage 3

Comment:

The patient's history of mantle cell lymphoma (MCL) s/p one cycle of treatment with Rituxan is noted. While no evidence of overt bone marrow involvement by MCL is noted, a small (1.0%) population of CD5+, kappa restricted B cells is detected by flow cytometric analysis, which is compatible with MCL and suspicious for low level involvement. This may be related to the patient's therapy. Clinical correlation is required.

Representative bone marrow aspirate samples were submitted to Integrated Oncology in Brentwood, TN. No molecular/cytogenetic studies were ordered at the time of biopsy. The bone marrow samples are held for 14 days for additional studies, if clinically indicated. They can be requested through send-outs (305-6771).

*** Report Electronically Signed ***

[redacted]

rss/12/30/2021 Yorke D. Young, M.D.

Addendum

ADDENDUM REPORT: Additional information added without a change in the diagnosis.
ADDENDUM NUMBER: 1
ADDENDUM DATE: 1/5/2022

ADDENDUM INFORMATION:

Per report, **no assay-specific abnormalities are detected by TCR (14q11.2) FISH probe**. Genetic changes other than those assayed cannot be ruled out on the basis of this testing. For more information, please see Integrated O[redacted] dated 1/3/2022.

*** Report Electronically Signed ***
Yorke D. Young, M.D.

Addendum

ADDENDUM REPORT: Additional information added without a change in the diagnosis.
ADDENDUM NUMBER: 1
ADDENDUM DATE: 1/5/2022

ADDENDUM INFORMATION:

Per report, fluorescence in situ hybridization (FISH) analysis utilizing DNA probes for abnormalities associated with mature B cell lymphomas (**Lymphoma FISH panel**) **detected no assay specific abnormalities**. Genetic changes other than those assayed cannot be ruled out on the basis of this testing. For more information please see Integrated Oncology report[redacted] dated 12/29/2021.

*** Report Electronically Signed ***
Yorke D. Young, M.D.

Gross Description

The specimen consists of a single small, cylindrically-shaped piece of cancellous bone measuring 1.7 cm in length. (cassette verified)

TLA/psp/12/29/2021

Microscopic Description

PERIPHERAL BLOOD:

A CBC performed on 12/29/21 discloses the following:

WBC	-	4.13 [k/uL]
RBC	-	3.75 [m/uL]
HGB	-	10.1 [g/dL]
HCT	-	30.7 [%]
MCV	-	81.9 [fL]

MCH	-	26.9 [pg]	
MCHC	-	32.9 [g/dL]	
RDW	-	17.7 [%]	
PLT	-	277 [k/uL]	
MPV	-	9.0 [fL]	
NEUT	-	2.82 [k/uL]	68.2 [%]
LYMPH	-	0.80 [k/uL]	19.4 [%]
MONO	-	0.23 [k/uL]	5.6 [%]
EO	-	0.25 [k/uL]	6.1 [%]
BASO	-	0.02 [k/uL]	0.5 [%]
IG	-	0.01 [k/uL]	0.2 [%]
NRBC	-	0.00 [k/uL]	0.0 [/100WBC]
RET	-	0.95 [%]	0.0356 [m/uL]
IRF	-	6.4 [%]	
RET-He	-	31.3 [pg]	
IPF	-	1.0 [%]	
RPI	-	0.49	

Peripheral blood smears and CBC data are reviewed. The peripheral blood is remarkable for a normocytic and normochromic anemia with occasional elliptocytes and no significant anisocytosis. There is no evidence of rouleaux, RBC agglutination, basophilic stippling, circulating nucleated red blood cells, or increased schistocytes. The reticulocyte production index is <2. The WBC is normal and all leukocytes are within normal absolute reference ranges. The lymphocytes are small to moderate size and mature without overt atypical features. Occasional reactive and large granular forms are noted. The granulocytes are appropriately segmented without overt dysplastic features. No circulating immature myeloid precursors or blasts are identified. The monocytes and eosinophils are morphologically mature. The platelets are quantitatively normal. No agranular or giant forms are identified. There is no significant platelet clumping. The immature platelet fraction is within normal limits.

BONE MARROW ASPIRATE:

The bone marrow aspirate smears are spicular, cellular, and adequate for evaluation. A 500 cell manual aspirate smear differential reveals: 0.8% blasts/blast equivalents, 1.5% promyelocytes, 9.6% myelocytes, 0.8% metamyelocytes, 19.6% bands, 16.2% segs, 8.0% eosinophils, 0.4% basophils, 5.0% lymphocytes, 0.6% monocytes, 1.0% plasma cells and 36.6 maturing erythroid precursors. The M:E ratio is slightly decreased at 1.3:1. Blasts are not increased. There is no overt dysplasia of the myeloid or erythroid lineages. Megakaryocytes display a spectrum of maturation and are evenly distributed. There is no overt lymphocytosis (focally prominent) or plasmacytosis.

BONE MARROW CORE BIOPSY AND CLOT / PARTICLE PREPARATION:

Evaluation of the bone marrow core biopsy reveals a normocellular marrow for age (30-40%) with trabeculae of normal thickness. Myeloids elements display adequate maturation and erythroid islands are present. The megakaryocytes are quantitatively normal without significant morphologic atypia or clustering. There are several small, well circumscribed, interstitial lymphoid aggregates comprised of small, mature cells. No overtly lymphoid aggregates or granulomas are seen. The bone marrow clot section displays similar findings.

FLOW CYTOMETRY:

Medical Necessity:	Evaluation for hematolymphoid neoplasia	
Viability:	80.1%	
Events per tube:	Adequate	
Abnormal B-cells:	1.0%	Phenotype: CD19+, CD20+, CD22+, CD5+(variable), CD10-, CD23-, CD38+, CD103-, FMC7+, Kappa+
Blasts:	0.7%	No significant phenotypic aberrancy

Lymphocytes:		
- B cells:	0.0%	
- NK cells:	1.1%	
- T cells:	8.3%	CD4:CD8 ratio: 0.8
Granulocytes:	74.4%	Normal maturation
Monocytes:	2.3%	Normal maturation
Eosinophils:	4.0%	
Basophils:	0.5%	
Plasma Cells:	0.2%	
Hematogones:	0.3%	
CD45- events/debris:	Remaining events	

Antibodies Performed: CD2, CD3, CD4, CD5, CD7, CD8, CD10, CD11b, CD11c, CD13, CD14, CD16, CD19, CD20, CD22, CD23, CD33, CD34, CD38, CD45, CD56, CD57, CD64, CD103, CD117, HLADR, FMC7, kappa, lambda

SPECIAL STAINS:

Stains are performed with adequate controls. The results are as follows:
Iron: Adequate storage iron (1+/4+) without ring sideroblasts
Reticulin: Focally increased within lymphoid aggregates, otherwise, not significantly increased

IMMUNOHISTOCHEMICAL STAINS:

Documentation of Medical Necessity for Immunohistochemistry: To evaluate lymphocyte distribution and clonality.

Description of Immunohistochemistry: Immunostains are performed with adequate positive and negative controls. The results are as follows: CD3 and CD20/CD79a highlight T-cells and B cells, respectively, scattered throughout the interstitium and mixed within the lymphoid aggregates. B cells slightly predominate within the aggregates. CD5 appears to only highlight the T-cells. PAX5 and CyclinD1 are appear negative, which may be technical artifact, possibly related to decalcification.

STAIN SUMMARY:
A1: CD3, CD20, CD5, CyclinD1, CD79a, PAX5, Reticulin
C1: Iron

In some tests, analyte specific reagents (ASRs) are used. In the case of an ASR, this test was developed and its performance characteristics determined by this laboratory. It has not been cleared or approved by the US Food and Drug Administration. The FDA has determined that such clearance or approval is not necessary. This test is used for clinical purposes. It should not be regarded as investigational or for research. This laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA '88) as qualified to perform high complexity clinical laboratory testing.

The technical component of the flow cytometric analysis was performed at Integrated Oncology in Brentwood, Tennessee. For more information; please see Integrated Oncology report BFT21-028798, dated 12/30/2021. The test was developed and its performance characteristics determined by LabCorp. It has not been cleared or approved by the U.S. Food and Drug Administration. The FDA has determined that such clearance or approval is not necessary. This test is used for clinical purposes. It should not be regarded as investigational or for research. This laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA-88) as qualified to perform high complexity clinical laboratory testing.